

BUSINESS ANALYTICS PROJECT

Business Analytics for Supermarket Customer and Sales Analysis

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Project documentation completed using the supplied project notebook and the supplied Applied Business Project format.

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5. Introduction

Instructions

Introduce the selected business domain and explain how Business Analytics can help the business.

- What is Business Analytics?
- Why is data important for businesses?
- What business area is selected?
- Why was this business selected?
- How can analytics help this business?

Project Answer

What is Business Analytics? Business Analytics is the use of data, statistics and software to understand business performance and support better decisions.

Why is data important? Data shows what customers buy, when they buy and how much they spend. Data-based decisions are more accurate than decisions based on guesses.

Business area selected: Retail – a supermarket chain.

Why this business? A supermarket produces many daily transactions, giving sufficient data for multiple analytics techniques.

How can analytics help? It can identify best-selling products, stronger branches, customer spending patterns, future sales trends and suitable stock levels.

6. Problem Statement

Instructions

Clearly describe the business problem that you are trying to solve. The problem statement should explain:

- What problem does the business face?
- What data is available?
- What needs to be analyzed?
- What decisions need to be taken?

Project Answer

To analyze supermarket sales data to identify customer purchasing patterns, factors affecting sales, future sales trends and suitable business strategies.

Problem: The supermarket needs to identify which products, branches and customer groups contribute most to sales.

Data available: 500 cleaned invoices from Jan–Mar 2026 containing product, branch, price, quantity, payment, rating and sales information.

Analysis required: Sales patterns, variable relationships, customer/payment behavior, future sales and business decision models.

Decisions: Product promotion, branch improvement, stock levels, marketing and payment-related strategies.

7. Objectives

Instructions

Write 5 to 8 clear objectives.

Project Answer

- To collect and clean supermarket sales data.
- To perform exploratory data analysis.
- To identify relationships among business variables.
- To predict future sales.
- To perform statistical testing.
- To analyze customer behaviour.
- To identify suitable business strategies.
- To support business decision-making using data.

8. Business Scenario / Domain Description

Instructions

Explain the selected business.

Include:

- Business name or hypothetical business name
- Type of business
- Products/services offered
- Target customers
- Business operations
- Important business problems

Students may use a real business or create a hypothetical business.

Project Answer

- **Business name:** FreshMart Supermarket (hypothetical).
- **Type:** Retail supermarket with Andheri, Borivali and Thane branches.
- **Products:** Groceries, Beverages, Snacks, Personal Care and Home Care.
- **Target customers:** Families and working professionals; Member and Normal customer types.
- **Operations:** Daily billing and stock management with Cash, Card and UPI payments.
- **Important problems:** Selecting stock levels, identifying high-value products and planning promotions.

9. Dataset Description

Instructions

Provide complete information about the dataset.

Mention:

- Dataset name
- Source
- Number of records
- Number of columns
- Description of each column
- Data types

- Numerical variables
- Categorical variables
- Date/time variables

Students should also attach the dataset or provide its source.

Project Answer

Dataset name: supermarket_sales.csv. **Source:** Sample supermarket sales data created for this project, using the same layout as public supermarket datasets. The notebook initially loads 503 rows including 3 duplicates; after preprocessing, 500 records are analyzed. **Columns:** 11.

Column	Description	Data Type
Invoice_ID	Unique invoice number	Categorical
Date	Date of purchase	Date/Time
Branch	Store branch	Categorical
Customer_Type	Member or Normal	Categorical
Gender	Customer gender	Categorical
Product_Line	Product category	Categorical
Unit_Price	Price of one unit (Rs.)	Numerical
Quantity	Units purchased	Numerical
Payment	Cash, Card or UPI	Categorical
Rating	Customer rating (4–10)	Numerical
Sales	Unit_Price × Quantity (Rs.)	Numerical

10. Data Collection

Instructions

Explain how the data was obtained.

Students can use:

- Survey data
- Public datasets
- Kaggle datasets
- Business-generated datasets
- College/department-approved datasets

Mention:

- Data source
- Data collection method
- Number of records collected
- Time period
- Variables collected

If survey data is used, attach the questionnaire.

Project Answer

Data source: business-generated/sample transaction data prepared for academic analysis. Collection method: point-of-sale style supermarket billing records. Final analyzed records: 500. Time period: 1 January 2026 to 31 March 2026. Variables: all 11 columns listed in the dataset description.

11. Data Preprocessing

Instructions

Show the steps performed before analysis. Students should check:

1. Missing values
2. Duplicate records
3. Incorrect data types
4. Invalid values
5. Outliers
6. Inconsistent values

Explain how each problem was handled.

Include relevant Python code and output screenshots.

Project Answer

Missing values: 5 missing Rating values were filled with the median.

Duplicates: 3 duplicate records were removed, changing the dataset from 503 to 500 records.

Data type: Date was converted to datetime format.

Invalid values: No invalid Unit_Price or Quantity records were found.

Outliers: 35 high-value Sales records were detected by the IQR method. They were retained because they represent genuine bulk purchases.

Inconsistent values: Text fields were stripped and standardized using title case. After cleaning, 0 missing values and 0 duplicate records remained.

12. Exploratory Data Analysis

Instructions

Perform EDA using Python. Students should:

- Display first five records.
- Display last five records.
- Display dataset shape.
- Display column names.
- Display data types.
- Check missing values.
- Check duplicate records.
- Find unique values.
- Calculate basic statistics.

Create suitable charts.

Minimum requirement:

- 2 bar charts
- 2 histograms
- 1 box plot
- 1 pie chart

- 1 scatter plot

Write observations below every important chart.

Project Answer

After preprocessing, the dataset shape is **(500, 11)**. There are 500 unique invoice IDs, 90 unique dates, 3 branches, 2 customer types, 2 genders, 5 product lines, 3 payment methods and 10 quantity values. No missing values or duplicates remain.

Charts performed: Sales by Product Line, Sales by Branch, Sales histogram, Rating histogram, Sales box plot, Payment Method pie chart and Quantity vs Sales scatter plot.

Observation: Personal Care and Home Care have the highest total sales, while Snacks has the lowest. Andheri and Borivali are close in sales, while Thane is lower. Sales is right-skewed and higher quantities generally produce higher sales. UPI is the most used payment method.

13. Descriptive Statistical Analysis

Instructions

Select important numerical variables and calculate:

- Mean
- Median
- Mode
- Minimum
- Maximum
- Range
- Variance
- Standard deviation

Present the results in a table.

Then explain what the values indicate about the business data.

Project Answer

Variable	Mean	Median	Mode	Minimum	Maximum	Range	Variance	Std Dev
Unit_Price	85.76	66.16	26.44	10.57	249.34	238.77	3608.07	60.07
Quantity	5.31	5.00	3.00	1	10	9	7.99	2.83
Sales	455.42	311.06	45.80	13.05	2283.30	2270.25	183007.26	427.79
Rating	7.15	7.20	8.50	4.00	10.00	6.00	2.81	1.68

Mean Sales (Rs. 455.42) is higher than median Sales (Rs. 311.06), showing that large invoices pull the average upward. Sales has a large standard deviation, indicating substantial variation in bill values. Average quantity is about 5.31 units per invoice.

14. Distribution Analysis

Instructions

Select at least one important numerical variable. Create:

- Histogram
- Box plot

Analyze:

- Shape of distribution
- Central tendency
- Spread
- Outliers
- Skewness

Write a short interpretation.

Project Answer

Sales was selected. Mean = Rs. 455.42, median = Rs. 311.06, standard deviation = Rs. 427.79, skewness = 1.64 and 35 IQR outliers were identified. The histogram is positively/right-skewed, with most bills relatively small and a smaller number of very large bills. The box plot confirms high-value observations on the upper side.

15. Correlation Analysis

Instructions

Select at least three numerical variables. Perform correlation analysis.

Create:

- Correlation matrix
- Heatmap

Identify:

- Positive relationships
- Negative relationships
- Strong relationships
- Weak relationships

Explain the important relationships in business terms.

Project Answer

	Unit_Price	Quantity	Sales	Rating
Unit_Price	1.00	0.00	0.72	0.06
Quantity	0.00	1.00	0.57	-0.05
Sales	0.72	0.57	1.00	0.01
Rating	0.06	-0.05	0.01	1.00

Sales has a strong positive relationship with Unit_Price ($r = 0.72$) and a moderate positive relationship with Quantity ($r = 0.57$). Rating has almost no relationship with Sales ($r = 0.01$). No notable negative relationship is present.

16. Regression Analysis

Instructions

Select:

- One independent variable
- One dependent variable

Perform linear regression.

Include:

- Regression equation

- Regression coefficient
- R^2 score
- Prediction
- MSE or RMSE
- Scatter plot
- Regression line

Explain the results.

Project Answer

Independent variable: Quantity. **Dependent variable:** Sales.

Regression equation: Sales = $-0.42 + 85.88 \times \text{Quantity}$.

Coefficient: 85.88. **R^2 :** 0.322. **MSE:** 123809.29. **RMSE:** 351.87.

Prediction: For 8 units, predicted sales = Rs. 686.60.

Interpretation: each additional unit is associated with about Rs. 85.88 higher sales. Quantity explains about 32.2% of sales variation; unit price and other factors explain the remaining variation.

17. Business Intelligence Dashboard

Instructions

Use Tableau or Power BI.

Create an interactive dashboard containing: • KPI cards • Bar chart • Pie chart • Line chart • Filters • Appropriate titles • Labels

Students should include screenshots of the dashboard. Write at least five insights obtained from the dashboard.

Project Answer

Dashboard data: dashboard_data.csv was prepared from the cleaned project dataset for use in Tableau or Power BI.

KPI cards: Total Sales, Number of Invoices, Average Rating. **Charts:** Sales by Product_Line bar chart, Sales by Payment pie chart, Sales by Date line chart. **Filters:** Branch, Customer_Type and Product_Line.

- Personal Care is the top product line by sales.
- Andheri and Borivali are the strongest branches.
- UPI is the most used payment method.
- January had the highest sales of the three months.
- Members spend more per bill than Normal customers.

The supplied project notebook contains the dashboard specifications and generates dashboard_data.csv; no actual Tableau/Power BI screenshot was embedded in the supplied notebook, so this report records the exact dashboard content rather than claiming a screenshot was supplied.

18. Probability and Probability Distributions

Instructions

Perform a simple probability experiment using Python. Example:

- Dice rolling
- Customer purchase outcome

- Product demand

Calculate probabilities of different outcomes. Generate random values using:

- Normal distribution
- Uniform distribution
- Exponential distribution

Create suitable graphs and explain their characteristics.

Project Answer

Experiment results: $P(\text{rolling a 6}) = 0.170$; $P(\text{even number}) = 0.496$; $P(\text{Customer is Member}) = 0.464$; $P(\text{Sales above Rs. 500}) = 0.342$; $P(\text{UPI payment}) = 0.454$.

Distributions: Normal distribution is approximately bell-shaped, Uniform is approximately flat across its range, and Exponential has many small values with a long right tail.

19. Decision Making Under Uncertainty

Instructions

Select a business decision involving uncertainty.

Identify:

- Alternatives
- Possible outcomes
- Probabilities
- Payoffs

Create a decision tree.

Calculate expected value for each alternative.

Compare the alternatives and explain the result.

Project Answer

Alternative	High (0.3)	Medium (0.5)	Low (0.2)	Expected Value (Rs. lakh)
Launch Premium Product	+80	+30	-40	31.0
Launch Budget Product	+50	+25	-10	25.5
Do Not Launch	0	0	0	0.0

Decision: Launch Premium Product because it has the highest expected value of Rs. 31 lakh. However, it also has the largest possible loss, while the Budget Product is safer.

20. Sampling and Sampling Distribution

Instructions

Select a sample from the dataset.

Mention:

- Population
- Sample
- Sample size
- Sampling technique

Take multiple random samples and calculate the sample mean. Repeat the process at least 100 times.

Create a histogram of sample means.

Calculate:

- Mean of sample means
- Standard deviation
- Standard error

Use different sample sizes and compare the results.

Explain the Central Limit Theorem.

Project Answer

Population: all 500 invoices. **Sample:** 30 invoices. **Technique:** simple random sampling. **Repeated:** 100 times.

For n = 30: mean of sample means = 448.87; SD of sample means = 84.75; reported standard error = 78.10.

Sample Size	Mean of Sample Means	SD of Sample Means
10	453.74	130.65
20	461.10	95.64
30	455.36	81.10
40	463.51	57.19

Central Limit Theorem: repeated random sample means form an approximately bell-shaped distribution around the population mean. As sample size increases, the spread of sample means decreases.

21. Confidence Interval Estimation

Instructions

Calculate confidence intervals for an important population parameter. Calculate:

- 90% confidence interval
- 95% confidence interval
- 99% confidence interval

Present the results in a table.

Explain the meaning of the 95% confidence interval in the selected business context.

Project Answer

Confidence Level	Lower Limit	Upper Limit
90%	370.45	499.51
95%	357.87	512.09
99%	332.91	537.05

Sample mean: Rs. 434.98. The 95% confidence interval indicates that, based on the selected sample and method, the true average sales per invoice is estimated to lie between Rs. 357.87 and Rs. 512.09. Higher confidence produces a wider interval.

22. Bootstrapping

Instructions

Select a numerical variable such as revenue or customer spending. Perform the following:

1. Create bootstrap samples.
2. Sample with replacement.

3. Calculate a statistic such as median.
4. Repeat at least 1,000 times.
5. Create a bootstrap distribution.
6. Calculate the 95% bootstrap confidence interval.
7. Interpret the result.

Include graph and output.

Project Answer

Variable selected: Sales. Original median = Rs. 311.06. After 1,000 bootstrap resamples, mean bootstrap median = Rs. 311.95 and the 95% bootstrap confidence interval is Rs. 281.03 to Rs. 341.00. This indicates that the median sales per invoice is centered close to Rs. 311.

23. Hypothesis Testing

Instructions

Identify one business question.

Formulate:

- Null hypothesis
- Alternative hypothesis
- Significance level

Perform an appropriate statistical test.

Report:

- Test statistic
- p-value
- Significance level
- Decision
- Business interpretation

Project Answer

Business question: Is the average sales per invoice different from Rs. 300?

H0: Mean sales per invoice = Rs. 300. **H1:** Mean sales per invoice $\neq$ Rs. 300. α : 0.05.

Test: One-sample t-test. Test statistic = 8.124; p-value $\approx$ 0.0000; decision = Reject H0.

Business interpretation: average sales per invoice is significantly different from Rs. 300 and is higher, with an observed mean of Rs. 455.42.

24. A/B Testing

Instructions

Select a website or marketing scenario.

Calculate conversion rates.

Formulate:

- Null hypothesis
- Alternative hypothesis

Perform an appropriate statistical test.

Compare the results and explain the business implication. Include a suitable chart.

Project Answer

Scenario: Version A = “Buy Now”; Version B = “Get Started”. 1,000 visitors were used for each version, with 120 conversions for A and 150 for B.

H0: Both buttons have the same conversion rate. **H1:** Conversion rates are different.

Results: A = 12%; B = 15%; $z = 1.963$; $p = 0.0496$; $\alpha = 0.05$; decision = Reject H0.

Business implication: Version B performs better, but the result is close to the significance threshold, so it should be tested for a longer period before making a permanent decision.

25. Time Series Analysis

Instructions

Use historical sales or demand data. The dataset should contain multiple time periods. Perform:

1. Convert date column into date/time format.
2. Set the date as the time index.
3. Create a time-series line chart.
4. Identify trend.
5. Identify seasonality if present.
6. Apply exponential smoothing.
7. Forecast future values.
8. Apply ARIMA.
9. Forecast future values.
10. Compare the forecasts.

Project Answer

Daily Sales was created by grouping Sales by Date and filling missing calendar dates with zero. A 7-day moving average was used for trend analysis. Average daily sales were highest on Sunday (Rs. 3360.68) and Saturday (Rs. 3090.61), while Wednesday was lowest (Rs. 1934.49), indicating a weekly pattern. January was the highest-sales month.

Forecast examples: Exponential Smoothing forecast $\approx$ Rs. 2240.43 for 18–22 March; ARIMA forecast $\approx$ Rs. 2503.41 to Rs. 2506.86 for the same early forecast dates. Both forecasts are relatively smooth.

26. Forecast Accuracy

Instructions

Evaluate the forecasting model using suitable measures such as: • MAE • MSE • RMSE

Present the results in a table.

Explain what the error values indicate.

Project Answer

Model	MAE	MSE	RMSE
Exponential Smoothing	1237.36	2322266.56	1523.90
ARIMA	1275.17	2240440.20	1496.81

Exponential Smoothing has the lower MAE, while ARIMA has the lower RMSE. Therefore, the preferred model depends on whether average absolute error or sensitivity to larger errors is prioritized.

27. Optimization Modeling

Instructions

Select a business optimization problem.

Define:

- Decision variables
- Objective function
- Constraints
- Non-negativity constraints

Solve the optimization problem using Python.

Report:

- Optimal solution
- Maximum profit/minimum cost
- Resource utilization

Explain the business meaning of the solution.

Project Answer

Decision variables: A and B = units of Product A and Product B. **Objective:** maximize Profit = $40A + 30B$.

Constraints: $2A + B \leq 100$ labour; $A + 2B \leq 80$ machine; $A, B \geq 0$.

Optimal solution: A = 40 units, B = 20 units. **Maximum profit:** Rs. 2,200. **Resource utilization:** 100/100 labour and 80/80 machine. Both resources are fully utilized.

28. Simulation Modeling

Instructions

Select a business scenario such as:

- Inventory management
- Customer arrivals
- Capacity planning
- Pricing
- Demand

Create a simulation using random values.

Compare at least three different scenarios.

For each scenario calculate suitable performance measures.

Create a comparison chart and explain the results.

Project Answer

Inventory simulation was performed for 365 days using random daily demand with an average of 40 units. Three stock levels were compared.

Scenario	Total Sales (units)	Shortage (units)	Service Level	Profit (Rs.)
Stock 35	12,500	2,017	86.1%	239,365

Scenario	Total Sales (units)	Shortage (units)	Service Level	Profit (Rs.)
Stock 45	14,439	273	98.1%	283,443
Stock 55	14,559	11	99.9%	280,093

Result: Stocking 45 units/day gives the highest profit and about 98.1% service level. Stocking 55 improves service only slightly but increases leftover inventory enough to reduce profit.

29. ANOVA

Instructions

Select a business problem involving three or more groups.

Formulate: H_0 : All group means are equal. H_1 : At least one group mean is different.

Perform one-way ANOVA.

Report:

- F-statistic
- p-value
- significance level
- Decision
- Business interpretation

Project Answer

Groups: Cash, Card and UPI payment methods. **H_0 :** Mean sales are equal across payment methods. **H_1 :** At least one mean differs.

Mean sales: Cash Rs. 468.19; Card Rs. 474.12; UPI Rs. 435.93. **F-statistic:** 0.437. **p-value:** 0.6463. α : 0.05.

Decision: Fail to reject H_0 .

Business interpretation: there is no statistically significant evidence that average spending differs by payment method.

30. Experimental Design

Instructions

Design an experiment to study the effect of different factors on a business outcome. Clearly identify:

- Factors
- Levels
- Response variable
- Experimental units

Explain how the experiment was conducted.

Project Answer

Factor 1: Advertising Type – Social Media, Email. **Factor 2:** Discount – 5%, 15%, 25%. **Response:** Average sales per store. **Experimental units:** 60 store-weeks, with 10 for each combination. Each store-week is randomly assigned an advertising type and discount level, and sales are recorded at the end of the week.

31. Factorial Experiment

Instructions

Select at least two factors.

Analyze:

- Main effect of Factor 1
- Main effect of Factor 2
- Interaction effect

Use suitable charts and statistical analysis.

Explain the results in business terms.

Project Answer

2 × 3 factorial experiment: Advertising Type (Social Media, Email) × Discount (5%, 15%, 25%), with Sales as the response.

Effect	F	p-value	Decision at 5%
Advertising Type	10.8556	0.0017	Significant
Discount	41.7916	<0.0001	Significant
Advertising × Discount	0.3190	0.7282	Not significant

Business interpretation: both advertising type and discount level significantly affect sales in the simulated experiment. The interaction is not significant, so the effect of discount does not materially depend on advertising type in this experiment.

32. Business Insights and Findings

Instructions

Combine the important findings from all analyses.

Write at least 8 to 10 meaningful business insights.

Each insight should be supported by:

- Statistical result
- Chart
- Model
- Table
- Prediction

Avoid writing general statements such as “sales are good.”

Project Answer

- **Personal Care and Home Care** have the highest sales; their average bills are much higher than Snacks.
- **Thane** has lower sales than Andheri and Borivali, indicating a branch-level improvement opportunity.
- **Members** spend more per bill than Normal customers.
- **UPI** is the most used payment method at 45.4%.
- Sales is right-skewed (skewness 1.64); mean Rs. 455.42 is above median Rs. 311.06.
- Unit_Price has strong positive correlation with Sales (0.72), while Quantity has moderate positive correlation (0.57).
- Regression estimates about Rs. 85.88 additional sales for each additional unit, with $R^2 = 0.322$.
- Average sales is significantly different from Rs. 300 ($t = 8.124$, $p < 0.05$).
- Payment method does not significantly affect spending (ANOVA $p = 0.6463$).
- Inventory simulation identifies 45 units/day as the most profitable stock level, with 98.1% service level.
- Version B “Get Started” has 15% conversion versus 12% for Version A, with $p = 0.0496$.

- Weekend sales are stronger than mid-week sales, especially Sunday and Saturday.

33. Business Recommendations

Instructions

Based on the analysis, provide 5 to 8 recommendations.

Recommendations should be directly related to your findings.

Project Answer

- Promote Personal Care and Home Care products because they generate high sales per bill.
- Run targeted promotions in Thane to improve its sales performance.
- Encourage customers to become Members because Members have higher average spending.
- Keep UPI as a major payment option and consider targeted UPI offers.
- Maintain around 45 units/day for fast-moving items in the simulated inventory scenario.
- Use sales forecasts for weekly stock planning and prepare additional stock for weekends.
- Use the “Get Started” button as the current better-performing option, while continuing A/B testing.

34. Conclusion

Instructions

Summarize the complete project. Mention:

- What problem was studied?
- What analysis was performed?
- What important findings were obtained?
- How can the business use these findings?
- What was learned from the project?

Project Answer

This project studied supermarket sales data to understand customer purchasing patterns, product and branch performance, sales relationships and future trends. The data was cleaned and analyzed using EDA, descriptive statistics, correlation, regression, probability, decision-making, sampling, confidence intervals, bootstrapping, hypothesis testing, A/B testing, time-series forecasting, optimization, simulation, ANOVA and factorial experimental design. The main findings are that Personal Care and Home Care drive sales, Thane needs improvement, Members spend more, quantity and unit price are important sales drivers, and a simulated stock level of about 45 units/day gives the best profit. These findings can support stock planning, promotions, branch strategy and forecasting. The project demonstrates how multiple Business Analytics techniques can work together to support data-driven decisions.

36. Appendix

Instructions

Include supporting material such as:

- Complete Python code
- Dataset sample
- Important outputs
- Additional graphs

- Tableau/Power BI screenshots
- Survey questionnaire
- Decision tree
- Optimization output
- Simulation output

Project Answer

Supporting material: The supplied project notebook *Business_Analytics_Project_Sagar_T010.ipynb* contains the complete Python workflow, dataset-generation/loading logic, preprocessing, EDA charts, statistical calculations, regression, probability distributions, decision tree logic, sampling, confidence intervals, bootstrapping, hypothesis testing, A/B testing, time-series forecasting, optimization, simulation, ANOVA and factorial experiment. The notebook also generates *dashboard_data.csv* for Tableau/Power BI.

Important outputs included in this report: descriptive-statistics table, correlation matrix, regression metrics, probability results, decision alternatives and expected values, sampling results, confidence intervals, bootstrap interval, hypothesis-test result, A/B-test result, forecast-accuracy table, optimization result, inventory simulation comparison, ANOVA result and factorial-experiment result.

Note: The supplied notebook did not contain an embedded Tableau/Power BI screenshot or a survey questionnaire; this report therefore does not falsely claim either item was supplied.